EC304 — Network Theory

Semester III • 3L:0T:0P • 3 Credits

Module 3 Exam Notes: Laplace Transforms, Network Transients & Initial Conditions

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

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1. Singularity Functions and Waveform Synthesis

Singularity functions (also referred to as switching functions or test signals) are a family of mathematical idealizations exhibiting discontinuities or discontinuous derivatives. In linear circuit theory, they model switching actions, sharp current/voltage pulses, and sudden transients.

1.1 Standard Singularity Functions

Definitions of Core Singularity Functions

1. **Unit Step Function** $u(t)$**:** Models the instantaneous closing or opening of an ideal switch at $t = 0$:

$$u(t) = \begin{cases} 0, & t < 0 \\ 1, & t > 0 \end{cases} \qquad \text{Delayed: } u(t - t_0) = \begin{cases} 0, & t < t_0 \\ 1, & t > t_0 \end{cases} \tag{1}$$

2. **Unit Impulse (Dirac Delta) Function** $\delta(t)$**:** Defined as the distributional derivative of the unit step function:

$$\delta(t) = \frac{du(t)}{dt}, \quad \delta(t) = 0 \text{ for } t \neq 0, \quad \int_{-\infty}^{\infty} \delta(t)\, dt = 1 \tag{2}$$

Sifting (Sampling) Property:

$$\int_{-\infty}^{\infty} f(t)\, \delta(t - t_0)\, dt = f(t_0) \tag{3}$$

3. **Unit Ramp Function** $r(t)$**:** Represents a linearly growing excitation starting at $t = 0$:

$$r(t) = \int_{-\infty}^{t} u(\tau)\, d\tau = t\, u(t) = \begin{cases} 0, & t < 0 \\ t, & t \geq 0 \end{cases} \tag{4}$$

4. **Unit Doublet** $\delta'(t)$**:** The time derivative of the unit impulse function:

$$\delta'(t) = \frac{d\delta(t)}{dt}, \qquad \int_{-\infty}^{\infty} f(t)\, \delta'(t - t_0)\, dt = -f'(t_0) \tag{5}$$

Hierarchy of Singularity Functions

The standard singularity functions form a continuous calculus chain linked by differentiation and integration:

$$\cdots \xleftarrow{\frac{d}{dt}} \delta'(t) \xleftarrow{\frac{d}{dt}} \delta(t) \xleftarrow{\frac{d}{dt}} u(t) \xleftarrow{\frac{d}{dt}} r(t) \xleftarrow{\frac{d}{dt}} \frac{t^2}{2} u(t) \cdots \tag{6}$$

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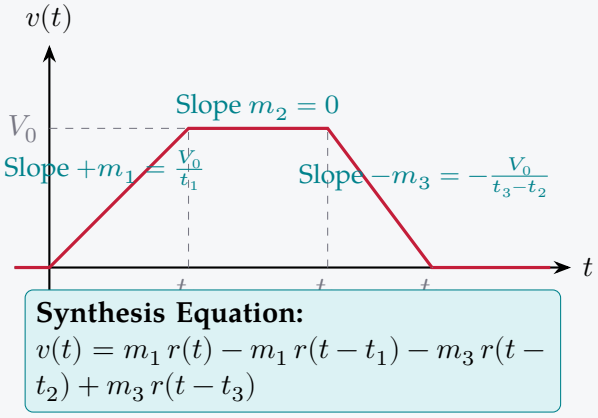
1.2 Waveform Synthesis Technique

Any arbitrary piecewise-linear or stepped signal can be synthesized systematically as a linear algebraic sum of delayed ramp functions and delayed step functions:

$$v(t) = \sum_{k=0}^{N} \Delta m_k \, r(t - t_k) + \sum_{j=0}^{M} \Delta v_j \, u(t - t_j) \tag{7}$$

where $\Delta m_k = m_k - m_{k-1}$ is the change in slope at breakpoint t_k, and Δv_j is the abrupt jump in amplitude at breakpoint t_j.

TikZ Schematic: Synthesis of a Trapezoidal Pulse from Singularity Functions



Synthesis Equation:
$v(t) = m_1 \, r(t) - m_1 \, r(t - t_1) - m_3 \, r(t - t_2) + m_3 \, r(t - t_3)$

Step-by-Step Example: Waveform Synthesis and its Transform

Problem: Express a triangular pulse that rises linearly from 0 to 10 V between $t = 0$ and $t = 2$ s, and drops instantaneously to zero at $t = 2$ s. Determine its Laplace transform.
Solution:

1. Between $t = 0$ and $t = 2$, the waveform has a constant slope $m_1 = \frac{10-0}{2-0} = +5$ V/s. It starts at $t = 0$, represented as $5\,r(t)$.
2. At $t = 2$ s, the ramp must be cancelled by subtracting an identical ramp of slope 5: $-5\,r(t-2)$.
3. At $t = 2$ s, the voltage must also jump downwards instantaneously by -10 V to reach 0: $-10\,u(t-2)$.
4. Synthesized time-domain expression:

$$v(t) = 5\,r(t) - 5\,r(t-2) - 10\,u(t-2) \tag{8}$$

5. Applying the time-shifting property of the Laplace transform:

$$V(s) = \frac{5}{s^2} - \frac{5}{s^2}\,e^{-2s} - \frac{10}{s}\,e^{-2s} = \frac{5}{s^2}\left[1 - e^{-2s}(1 + 2s)\right] \tag{9}$$

2. The Laplace Transform and Operational Theorems

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The unilateral (one-sided) Laplace transform is an essential mathematical tool for solving linear, time-invariant (LTI) differential equations with non-zero initial boundary conditions.

2.1 Definition and Region of Convergence

Unilateral Laplace Transform

Let $f(t)$ be a continuous or piecewise-continuous function defined for $t \geq 0$. The unilateral Laplace transform $\mathcal{L}\{f(t)\} = F(s)$ is:

$$F(s) = \int_{0^-}^{\infty} f(t)\, e^{-st}\, dt \tag{10}$$

where $s = \sigma + j\omega$ is the complex frequency variable. The lower integration limit $0^- \equiv \lim_{\epsilon \to 0^+}(0 - \epsilon)$ ensures that singularity functions (such as $\delta(t)$) acting at $t = 0$ are rigorously integrated and accounted for.

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2.2 Operational Theorems for Circuit Analysis

Property / Theorem	Time Domain $f(t)$	Frequency Domain $F(s)$
Linearity	$a_1 f_1(t) + a_2 f_2(t)$	$a_1 F_1(s) + a_2 F_2(s)$
First Shifting (Frequency)	$e^{-at} f(t)$	$F(s+a)$
Second Shifting (Time)	$f(t-t_0)\, u(t-t_0)$	$e^{-st_0} F(s) \quad (t_0 \geq 0)$
Time Differentiation (1st)	$\dfrac{df(t)}{dt}$	$sF(s) - f(0^-)$
Time Differentiation (2nd)	$\dfrac{d^2 f(t)}{dt^2}$	$s^2 F(s) - sf(0^-) - f'(0^-)$
Time Integration	$\displaystyle\int_{0^-}^{t} f(\tau)\, d\tau$	$\dfrac{F(s)}{s}$
Frequency Differentiation	$t\, f(t)$	$-\dfrac{d}{ds} F(s)$
Time Convolution	$f_1(t) * f_2(t)$	$F_1(s) \cdot F_2(s)$
Initial Value Theorem	$\lim\limits_{t \to 0^+} f(t)$	$\lim\limits_{s \to \infty} sF(s) \quad$ (if limit exists)
Final Value Theorem	$\lim\limits_{t \to \infty} f(t)$	$\lim\limits_{s \to 0} sF(s) \quad$ (poles of $sF(s)$ in LHP)
Periodic Function (T)	$f(t) = f(t+T)$	$\dfrac{1}{1-e^{-sT}} \displaystyle\int_0^T f(t) e^{-st}\, dt$

Critical Examination Note: Validity of the Final Value Theorem (FVT)

The Final Value Theorem $\lim_{t\to\infty} f(t) = \lim_{s\to 0} sF(s)$ is valid **if and only if** the function $f(t)$ possesses a finite, stable steady-state limit as $t \to \infty$. Mathematically, this requires that all poles of $sF(s)$ reside strictly in the open left-half of the s-plane ($\text{Re}(s) < 0$).

- For $f(t) = \sin(\omega_0 t)$, $F(s) = \frac{\omega_0}{s^2+\omega_0^2}$. Here $sF(s) = \frac{s\omega_0}{s^2+\omega_0^2}$. Evaluating $\lim_{s\to 0} sF(s) = 0$, but $\lim_{t\to\infty} \sin(\omega_0 t)$ does not exist (it oscillates indefinitely). FVT fails because poles are on the $j\omega$-axis!

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2.3 Standard Laplace Transform Pairs

Signal $f(t)$	$F(s)$	**Signal** $f(t)$	$F(s)$
$\delta(t)$	1	$u(t)$	$\frac{1}{s}$
$t\,u(t)$	$\frac{1}{s^2}$	$t^n\,u(t)$	$\frac{n!}{s^{n+1}}$
$e^{-at}\,u(t)$	$\frac{1}{s+a}$	$t\,e^{-at}\,u(t)$	$\frac{1}{(s+a)^2}$
$\sin(\omega t)\,u(t)$	$\frac{\omega}{s^2+\omega^2}$	$\cos(\omega t)\,u(t)$	$\frac{s}{s^2+\omega^2}$
$e^{-at}\sin(\omega t)\,u(t)$	$\frac{\omega}{(s+a)^2+\omega^2}$	$e^{-at}\cos(\omega t)\,u(t)$	$\frac{s+a}{(s+a)^2+\omega^2}$

3. Inverse Laplace Transform and Partial Fraction Expansion

In network transient problems, the s-domain response is generally obtained as a rational transfer function $F(s) = \frac{P(s)}{Q(s)}$, where $P(s)$ and $Q(s)$ are polynomials in s with real coefficients, and $\deg(P) < \deg(Q)$ (strictly proper).

3.1 Decomposition Methodologies

Three Cases of Partial Fraction Expansion

1. **Case I: Distinct Real Roots (Simple Poles)**

$$F(s) = \frac{P(s)}{(s-p_1)(s-p_2)\cdots(s-p_n)} = \sum_{k=1}^{n} \frac{A_k}{s-p_k} \tag{11}$$

The residues A_k are evaluated by Heaviside's cover-up rule:

$$A_k = \lim_{s\to p_k} (s-p_k)\,F(s) \implies f(t) = \sum_{k=1}^{n} A_k\, e^{p_k t}\, u(t) \tag{12}$$

2. **Case II: Repeated Real Roots (Multiple Poles of Order m)**

$$F(s) = \frac{P(s)}{(s-p)^m\, Q_1(s)} = \frac{B_m}{(s-p)^m} + \frac{B_{m-1}}{(s-p)^{m-1}} + \cdots + \frac{B_1}{s-p} + \cdots \tag{13}$$

where the coefficients are given by:

$$B_{m-k} = \frac{1}{k!} \lim_{s\to p} \frac{d^k}{ds^k} \left[(s-p)^m\, F(s)\right], \quad k = 0, 1, \ldots, m-1 \tag{14}$$

3. **Case III: Complex Conjugate Poles ($s = -\alpha \pm j\beta$)**

$$F(s) = \frac{Cs + D}{(s+\alpha)^2 + \beta^2} = \frac{C(s+\alpha) + (D - C\alpha)}{(s+\alpha)^2 + \beta^2} \tag{15}$$

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The inverse transform yields a damped sinusoid:

$$f(t) = Ce^{-\alpha t}\cos(\beta t) + \frac{D - C\alpha}{\beta}e^{-\alpha t}\sin(\beta t) = Me^{-\alpha t}\cos(\beta t - \theta) \tag{16}$$

where $M = \sqrt{C^2 + \left(\frac{D-C\alpha}{\beta}\right)^2}$ and $\theta = \tan^{-1}\left(\frac{D-C\alpha}{\beta C}\right)$.

4. Evaluation of Initial Conditions in Electrical Networks

Evaluating initial conditions at the switching instant ($t = 0^-$ and $t = 0^+$) is an absolute prerequisite for setting up transient differential equations and their Laplace counterparts.

4.1 Physical Boundary and Continuity Principles

Energy Continuity Postulates

1. **Conservation of Inductor Flux Linkage:** An inductor stores magnetic energy $W_m = \frac{1}{2}Li_L^2$. An instantaneous discontinuity in $i_L(t)$ requires infinite voltage ($v_L = L\frac{di}{dt} \to \infty$) and infinite instantaneous power, which is physically impossible in passive circuits without impulse voltage sources. Therefore:

$$i_L(0^+) = i_L(0^-) \tag{17}$$

2. **Conservation of Capacitor Electric Charge:** A capacitor stores electrostatic energy $W_e = \frac{1}{2}Cv_C^2$. An instantaneous discontinuity in $v_C(t)$ requires infinite current ($i_C = C\frac{dv_C}{dt} \to \infty$), which is physically impossible in passive circuits without impulse current sources. Therefore:

$$v_C(0^+) = v_C(0^-) \tag{18}$$

4.2 Equivalent Circuits at $t = 0^-$ and $t = 0^+$

Element	DC Steady State ($t = 0^-$)	At $t = 0^+$ (with initial state)	At $t = 0^+$ (zero initial state)
Resistor (R)	**Resistor** R ($v_R = i_R R$)	**Resistor** R ($v_R = i_R R$)	Resistor R ($v_R = i_R R$)
Inductor (L)	**Short Circuit** ($v_L = 0$)	**Current Source** $I_0 = i_L(0^-)$	Open Circuit ($i_L(0^+) = 0$)
Capacitor (C)	**Open Circuit** ($i_C = 0$)	**Voltage Source** $V_0 = v_C(0^-)$	Short Circuit ($v_C(0^+) = 0$)

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4.3 Evaluation of Higher Derivatives at $t = 0^+$

To evaluate higher-order derivatives such as $\frac{di}{dt}(0^+)$, $\frac{dv_C}{dt}(0^+)$, and $\frac{d^2i}{dt^2}(0^+)$:

1. Write the loop KVL and nodal KCL equations for $t > 0$ containing derivative terms:

$$v_L(t) = L\,\frac{di_L(t)}{dt} \implies \left.\frac{di_L}{dt}\right|_{t=0^+} = \frac{v_L(0^+)}{L} \tag{19}$$

$$i_C(t) = C\,\frac{dv_C(t)}{dt} \implies \left.\frac{dv_C}{dt}\right|_{t=0^+} = \frac{i_C(0^+)}{C} \tag{20}$$

2. Calculate $v_L(0^+)$ and $i_C(0^+)$ directly by evaluating the network loop/node equations at the instant $t = 0^+$ using the known boundary values $i_L(0^+)$ and $v_C(0^+)$.

3. Differentiate the general differential equation with respect to t, and evaluate the resulting relation at $t = 0^+$ to find the second derivative $\frac{d^2i}{dt^2}(0^+)$.

Worked Example: Complete Evaluation of Initial Boundary Derivatives

Problem: In a series RLC circuit ($R = 10\,\Omega, L = 1\text{ H}, C = 0.1\text{ F}$), a DC voltage $V = 100$ V is suddenly applied at $t = 0$ by closing a switch. The circuit had no initial energy ($i_L(0^-) = 0, v_C(0^-) = 0$). Find $i(0^+)$, $\frac{di}{dt}(0^+)$, and $\frac{d^2i}{dt^2}(0^+)$.
Solution:

1. **At $t = 0^+$:** Due to continuity of inductor current, $i(0^+) = i_L(0^-) = 0$ A. Due to continuity of capacitor voltage, $v_C(0^+) = v_C(0^-) = 0$ V.

2. **First derivative $\frac{di}{dt}(0^+)$:** Applying KVL around the loop for $t \geq 0$:

$$R\,i(t) + L\,\frac{di(t)}{dt} + v_C(t) = V \tag{21}$$

Evaluating this at $t = 0^+$:

$$R\,i(0^+) + L\left.\frac{di}{dt}\right|_{t=0^+} + v_C(0^+) = V \implies 10(0) + 1\cdot\left.\frac{di}{dt}\right|_{t=0^+} + 0 = 100 \tag{22}$$

$$\left.\frac{di}{dt}\right|_{t=0^+} = \frac{100 - 0 - 0}{1} = 100\text{ A/s} \tag{23}$$

3. **Second derivative $\frac{d^2i}{dt^2}(0^+)$:** Differentiating the KVL equation with respect to time t:

$$R\,\frac{di(t)}{dt} + L\,\frac{d^2i(t)}{dt^2} + \frac{dv_C(t)}{dt} = 0 \implies R\,\frac{di}{dt} + L\,\frac{d^2i}{dt^2} + \frac{i(t)}{C} = 0 \tag{24}$$

Evaluating at $t = 0^+$:

$$10(100) + 1\cdot\left.\frac{d^2i}{dt^2}\right|_{t=0^+} + \frac{0}{0.1} = 0 \implies \left.\frac{d^2i}{dt^2}\right|_{t=0^+} = -1000\text{ A/s}^2 \tag{25}$$

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5. Transformed s-Domain Circuit Models

Instead of formulating differential equations in the time domain and then taking the transform, we can directly map each physical circuit element into its transformed algebraic s-domain model containing initial conditions.

5.1 Derivation of Element Equivalent Circuits

Mathematical Derivation of s-Domain Models

1. **Resistor (R):**

$$v(t) = R\,i(t) \iff V(s) = R\,I(s) \qquad \text{Impedance: } Z_R(s) = R \tag{26}$$

2. **Inductor (L):**

$$v_L(t) = L\,\frac{di_L}{dt} \iff V_L(s) = sL\,I_L(s) - L\,i_L(0^-) \tag{27}$$

- **Thevenin Form (Series):** An impedance sL in series with an opposing voltage source of magnitude $L\,i_L(0^-)$ (positive terminal at the current entry).
- **Norton Form (Parallel):** Solving for $I_L(s)$:

$$I_L(s) = \frac{V_L(s)}{sL} + \frac{i_L(0^-)}{s} \tag{28}$$

An admittance $\frac{1}{sL}$ in parallel with an independent current source $\frac{i_L(0^-)}{s}$ pointing in the direction of $i_L(t)$.

3. **Capacitor (C):**

$$i_C(t) = C\,\frac{dv_C}{dt} \iff I_C(s) = sC\,V_C(s) - C\,v_C(0^-) \tag{29}$$

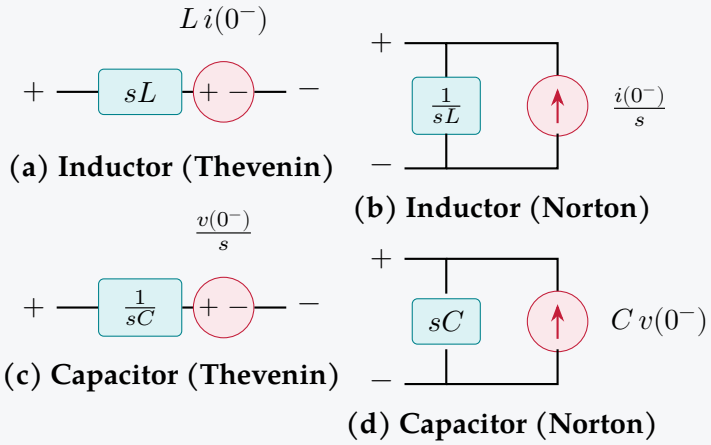
- **Norton Form (Parallel):** An admittance sC in parallel with an independent current source $C\,v_C(0^-)$ directed towards the positive terminal.
- **Thevenin Form (Series):** Solving for $V_C(s)$:

$$V_C(s) = \frac{I_C(s)}{sC} + \frac{v_C(0^-)}{s} \tag{30}$$

An impedance $\frac{1}{sC}$ in series with a voltage source $\frac{v_C(0^-)}{s}$ maintaining the polarity of $v_C(0^-)$.

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TikZ Schematic: Transformed s-Domain Circuit Equivalents with Initial Energy



(a) Inductor (Thevenin) (b) Inductor (Norton)

(c) Capacitor (Thevenin) (d) Capacitor (Norton)

6. Transient Analysis of First-Order Networks (RL and RC)

First-order circuits contain a single equivalent energy-storage element (L or C) and are mathematically governed by linear first-order ordinary differential equations.

6.1 Series RL Network with DC Step Voltage

Consider a series connection of resistance R and inductance L excited at $t = 0$ by a step voltage $V\,u(t)$ with an initial current $i(0^-) = I_0$.

Mathematical Derivation of RL Response

Applying KVL in the transformed s-domain:

$$\frac{V}{s} = R\,I(s) + sL\,I(s) - L\,I_0 \implies I(s)(R+sL) = \frac{V}{s} + L\,I_0 \tag{31}$$

Solving for the current transform $I(s)$:

$$I(s) = \frac{V}{s(sL+R)} + \frac{L\,I_0}{sL+R} = \frac{V/L}{s\left(s+\frac{R}{L}\right)} + \frac{I_0}{s+\frac{R}{L}} \tag{32}$$

Decomposing the first term via partial fractions:

$$\frac{V/L}{s\left(s+\frac{R}{L}\right)} = \frac{V/R}{s} - \frac{V/R}{s+\frac{R}{L}} \tag{33}$$

Substituting back and collecting terms:

$$I(s) = \frac{V/R}{s} - \frac{V/R}{s+\frac{R}{L}} + \frac{I_0}{s+\frac{R}{L}} \tag{34}$$

Taking the inverse Laplace transform for $t \geq 0$:

$$i(t) = \frac{V}{R}\left(1 - e^{-\frac{R}{L}t}\right) + I_0\,e^{-\frac{R}{L}t} = \underbrace{\frac{V}{R}}_{\text{Forced (SS)}} + \underbrace{\left(I_0 - \frac{V}{R}\right)e^{-t/\tau}}_{\text{Natural (Transient)}} \tag{35}$$

where $\tau = \frac{L}{R}$ is the inductive time constant (in seconds).

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Decomposition into Zero-Input and Zero-State Responses

- **Zero-Input Response (ZIR):** Response due entirely to initial internal stored energy ($V = 0$):

$$i_{\text{ZIR}}(t) = I_0\, e^{-t/\tau} \tag{36}$$

- **Zero-State Response (ZSR):** Response due entirely to external applied excitation with zero initial state ($I_0 = 0$):

$$i_{\text{ZSR}}(t) = \frac{V}{R}\left(1 - e^{-t/\tau}\right) \tag{37}$$

- **Principle of Linearity:** Total Response = ZIR + ZSR ≡ Natural + Forced.

6.2 Series RC Network with DC Step Voltage

For a series RC circuit excited by $V\,u(t)$ with initial capacitor voltage $v_C(0^-) = V_0$:

$$\frac{V}{s} = R\,I(s) + \frac{I(s)}{sC} + \frac{V_0}{s} \implies I(s)\left(R + \frac{1}{sC}\right) = \frac{V - V_0}{s} \tag{38}$$

Solving for loop current transform $I(s)$ and capacitor voltage transform $V_C(s)$:

$$I(s) = \frac{V - V_0}{R\left(s + \frac{1}{RC}\right)} \implies i(t) = \frac{V - V_0}{R}\, e^{-t/\tau} \quad (t \geq 0) \tag{39}$$

$$V_C(s) = \frac{V/RC}{s\left(s + \frac{1}{RC}\right)} + \frac{V_0}{s + \frac{1}{RC}} \implies v_C(t) = V\left(1 - e^{-t/\tau}\right) + V_0\, e^{-t/\tau} \tag{40}$$

where $\tau = RC$ is the capacitive time constant.

7. Transient Analysis of Second-Order Networks (RLC)

Second-order circuits contain two uncoupled energy storage elements (L and C) whose transient response is described by a second-order linear differential equation.

7.1 Series RLC Circuit Formulation

Applying KVL to a series RLC circuit excited by a DC voltage $V\,u(t)$ with zero initial conditions:

$$R\,i(t) + L\,\frac{di(t)}{dt} + \frac{1}{C}\int_0^t i(\tau)\,d\tau = V \tag{41}$$

Taking the Laplace transform with zero initial conditions:

$$I(s)\left[R + sL + \frac{1}{sC}\right] = \frac{V}{s} \implies I(s)\left[\frac{s^2L + sR + \frac{1}{C}}{s}\right] = \frac{V}{s} \tag{42}$$

$$I(s) = \frac{V/L}{s^2 + \frac{R}{L}s + \frac{1}{LC}} = \frac{V/L}{s^2 + 2\alpha s + \omega_0^2} \tag{43}$$

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Standard Second-Order Characteristic Parameters

$$s^2 + 2\alpha s + \omega_0^2 = 0 \quad \text{or} \quad s^2 + 2\zeta\omega_0 s + \omega_0^2 = 0 \tag{44}$$

- **Neper Frequency (Attenuation Factor):** $\alpha = \dfrac{R}{2L}$ (rad/s)
- **Undamped Resonant Frequency:** $\omega_0 = \dfrac{1}{\sqrt{LC}}$ (rad/s)
- **Damping Ratio:** $\zeta = \dfrac{\alpha}{\omega_0} = \dfrac{R}{2}\sqrt{\dfrac{C}{L}}$

Roots of the characteristic polynomial:

$$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2} = -\zeta\omega_0 \pm \omega_0\sqrt{\zeta^2 - 1} \tag{45}$$

7.2 Classification of Damping Regimes

The nature of the transient response depends strictly on the discriminant $(\alpha^2 - \omega_0^2)$ or damping ratio ζ:

Three Classical Damping Regimes

1. **Overdamped** ($\zeta > 1 \iff R > 2\sqrt{L/C} \iff \alpha > \omega_0$)**:** The roots s_1, s_2 are real, negative, and unequal.

$$i(t) = \frac{V}{2\beta L}\left(e^{-(\alpha-\beta)t} - e^{-(\alpha+\beta)t}\right), \quad \text{where } \beta = \sqrt{\alpha^2 - \omega_0^2} \tag{46}$$

   The current rises to a peak and decays sluggishly to zero without any oscillation.

2. **Critically Damped** ($\zeta = 1 \iff R = 2\sqrt{L/C} \iff \alpha = \omega_0$)**:** The roots are real, negative, and identical: $s_1 = s_2 = -\alpha$.

$$i(t) = \frac{V}{L}\, t\, e^{-\alpha t} \tag{47}$$

   Provides the fastest non-oscillatory return to steady-state equilibrium.

3. **Underdamped** ($\zeta < 1 \iff R < 2\sqrt{L/C} \iff \alpha < \omega_0$)**:** The roots are a pair of complex conjugates in the left-half s-plane:

$$s_{1,2} = -\alpha \pm j\omega_d, \quad \text{where } \omega_d = \sqrt{\omega_0^2 - \alpha^2} = \omega_0\sqrt{1 - \zeta^2} \tag{48}$$

   ω_d is the damped natural frequency. The loop current is an exponentially decaying sinusoidal oscillation:

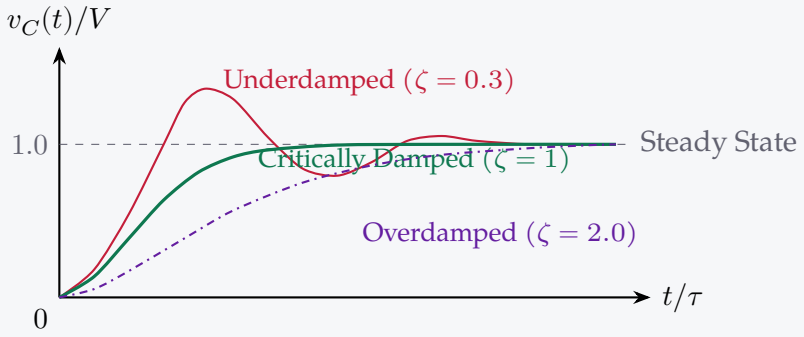
$$i(t) = \frac{V}{\omega_d L}\, e^{-\alpha t} \sin(\omega_d t) \tag{49}$$

   The capacitor voltage exhibits oscillatory overshoot:

$$v_C(t) = V\left[1 - \frac{e^{-\alpha t}}{\sqrt{1-\zeta^2}} \sin(\omega_d t + \phi)\right], \quad \phi = \cos^{-1}(\zeta) \tag{50}$$

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TikZ Plot: Comparison of Second-Order Step Responses for Different Damping Ratios



7.3 Dual Parallel RLC Network

In a parallel RLC circuit excited by an independent current step $I\,u(t)$, the dual nodal formulation gives:

$$\frac{v(t)}{R} + \frac{1}{L}\int_0^t v(\tau)\,d\tau + C\,\frac{dv(t)}{dt} = I \tag{51}$$

The characteristic equation and parameters are:

$$s^2 + \frac{1}{RC}s + \frac{1}{LC} = 0 \implies \alpha = \frac{1}{2RC}, \quad \omega_0 = \frac{1}{\sqrt{LC}}, \quad \zeta = \frac{1}{2R}\sqrt{\frac{L}{C}} \tag{52}$$

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Quick Revision — Module 3

Laplace Transforms, Network Transients and Initial Conditions | EC304

Key Formulas and Network Relations at a Glance

Unilateral Laplace Transform	$F(s) = \int_{0^-}^{\infty} f(t)\, e^{-st}\, dt, \quad s = \sigma + j\omega$
Time Derivative Transform	$\mathcal{L}\{f'(t)\} = sF(s) - f(0^-), \quad \mathcal{L}\{f''(t)\} = s^2F(s) - sf(0^-) - f'(0^-)$
Time Shifting Property	$\mathcal{L}\{f(t-t_0)u(t-t_0)\} = e^{-st_0}F(s) \quad (t_0 \geq 0)$
Final Value Theorem	$\lim_{t\to\infty} f(t) = \lim_{s\to 0} sF(s)$ (poles in open LHP)
Inductor s-Domain Model	$V_L(s) = sLI_L(s) - Li_L(0^-) \iff I_L(s) = \frac{V_L(s)}{sL} + \frac{i_L(0^-)}{s}$
Capacitor s-Domain Model	$V_C(s) = \frac{I_C(s)}{sC} + \frac{v_C(0^-)}{s} \iff I_C(s) = sCV_C(s) - Cv_C(0^-)$
First-Order Step Response	$x(t) = x(\infty) + [x(0^+) - x(\infty)]e^{-t/\tau}, \quad \tau_{RL} = \frac{L}{R}, \ \tau_{RC} = RC$
Series RLC Damping Parameters	$\alpha = \frac{R}{2L}, \quad \omega_0 = \frac{1}{\sqrt{LC}}, \quad \zeta = \frac{\alpha}{\omega_0} = \frac{R}{2}\sqrt{\frac{C}{L}}$
Damped Frequency	$\omega_d = \sqrt{\omega_0^2 - \alpha^2} = \omega_0\sqrt{1-\zeta^2} \quad (\text{for } \zeta < 1)$

One-Line Definitions

- **Singularity Function:** A function exhibiting discontinuities in its value or its higher-order time derivatives.
- **Unit Impulse $\delta(t)$:** A generalized function of infinite height, zero width, and unit integral area at $t = 0$.
- **Lower Limit 0^-:** The instant immediately prior to switching, capturing jump discontinuities and initial states.
- **Zero-Input Response (ZIR):** System response produced solely by initial stored energy with external sources zeroed.
- **Zero-State Response (ZSR):** System response produced solely by applied excitations with zero initial stored energy.
- **Time Constant (τ):** Time required for a first-order transient response to decay to 36.8% $(1/e)$ of initial value.
- **Neper Frequency (α):** Exponential damping attenuation factor governing the rate of response decay.
- **Undamped Resonant Frequency (ω_0):** Natural oscillation frequency of an LC circuit in the absence of resistance.
- **Damping Ratio (ζ):** Dimensionless ratio α/ω_0 determining whether a second-order circuit oscillates.
- **Critically Damped Circuit:** Second-order circuit with $\zeta = 1$ returning to steady state in minimum non-oscillatory time.

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Frequently Asked / Viva Questions

Q1. Why cannot inductor current change instantaneously in a finite circuit?

Stored magnetic energy is $W = \frac{1}{2}Li_L^2$. An instantaneous change requires infinite power ($P = \frac{dW}{dt} \to \infty$) and infinite voltage ($v_L = L\frac{di}{dt} \to \infty$), which is impossible with finite sources. Hence, $i_L(0^+) = i_L(0^-)$.

Q2. Why does an uncharged capacitor act as a short circuit at $t = 0^+$?

Since $v_C(0^+) = v_C(0^-) = 0\,\text{V}$, the terminal voltage is zero immediately after switching while current flows freely. An element supporting finite current with zero voltage drop behaves identically to an ideal short circuit.

Q3. Explain the physical necessity of the lower limit 0^- in unilateral Laplace transforms.

The 0^- limit captures Dirac delta impulses $\delta(t)$ acting at $t = 0$ and accounts for pre-switching initial conditions ($i_L(0^-), v_C(0^-)$). A lower limit of 0^+ would omit impulses occurring at the switching origin.

Q4. State the validity condition for applying the Final Value Theorem.

$\lim_{t\to\infty} f(t) = \lim_{s\to 0} sF(s)$ is valid strictly if all poles of $sF(s)$ lie in the open left-half s-plane. If poles lie on the imaginary axis (e.g. sustained sinusoids) or in the right-half plane, FVT produces incorrect results.

Q5. Distinguish between natural and forced responses in circuit transients.

The natural (transient) response depends solely on circuit storage elements (R, L, C) and decays to zero. The forced (steady-state) response is sustained by external driving sources and mirrors their functional form.

Q6. Under what switching condition is an AC RL circuit completely transient-free?

In an RL circuit excited by $v(t) = V_m \sin(\omega t + \theta)$, transient-free operation occurs when switching at $\theta - \phi = 0$ or π (where $\phi = \tan^{-1}\frac{\omega L}{R}$), matching prospective steady-state current to the initial zero current.

Q7. How does the Norton s-domain model of an inductor differ from its Thevenin form?

The Thevenin form consists of an impedance sL in series with an opposing voltage source $Li_L(0^-)$. The Norton form comprises an admittance $\frac{1}{sL}$ in parallel with a current source $\frac{i_L(0^-)}{s}$ pointing in the direction of i_L.

Q8. Define damping ratio ζ and give its formulas for series and parallel RLC circuits.

$\zeta = \alpha/\omega_0$. For series RLC: $\zeta = \frac{R}{2}\sqrt{\frac{C}{L}}$. For parallel RLC: $\zeta = \frac{1}{2R}\sqrt{\frac{L}{C}}$.

Q9. Why is critical damping ($\zeta = 1$) preferred in analog meters and galvanometers?

Critical damping ensures the pointer reaches its steady deflection in the shortest possible time without oscillatory overshoot or sluggish delay, enabling fast, accurate readings.

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Frequently Asked / Viva Questions (contd.)

Q10. What is the circuit significance of the unit doublet $\delta'(t)$?

The unit doublet represents $\frac{d\delta}{dt}$. In circuits, it arises from impulsive current injection across inductors or impulse voltage across capacitors, abruptly altering state variables at $t = 0$.

Summary Matrix: Comprehensive Comparison of First- and Second-Order Transients

Circuit Type	Time Const. / Damping	Characteristic Roots	Transient Response Characteri
Series RL	$\tau = \frac{L}{R}$	$s = -\frac{R}{L}$	Monotonic exponential rise decay; no oscillat
Series RC	$\tau = RC$	$s = -\frac{1}{RC}$	Monotonic exponential volta change; current jump at $t = 0^+$.
Series RLC Overdamped	$\alpha = \frac{R}{2L} > \omega_0,\ \zeta > 1$	$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$	Sum of two decay exponentials; sluggish, non-oscillatory.
Series RLC Critically Damped	$\alpha = \omega_0,\ \zeta = 1$	$s_1 = s_2 = -\alpha$	Rapid monotonic return to equilibrium with overshoot $((A_1 + A_2 t)e^{-\alpha t})$
Series RLC Underdamped	$\alpha < \omega_0,\ \zeta < 1$	$s_{1,2} = -\alpha \pm j\omega_d$	Damped sinusoic oscillations with decaying envelop $e^{-\alpha t}$.
Parallel RLC	$\alpha = \frac{1}{2RC},\ \zeta = \frac{1}{2R}\sqrt{\frac{L}{C}}$	$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$	Dual of series RL higher R produce lower damping (oscillatory).

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